Exp 34:Create a neural network using a popular deep learning library like TensorFlow or PyTorch for image classification.

Code:

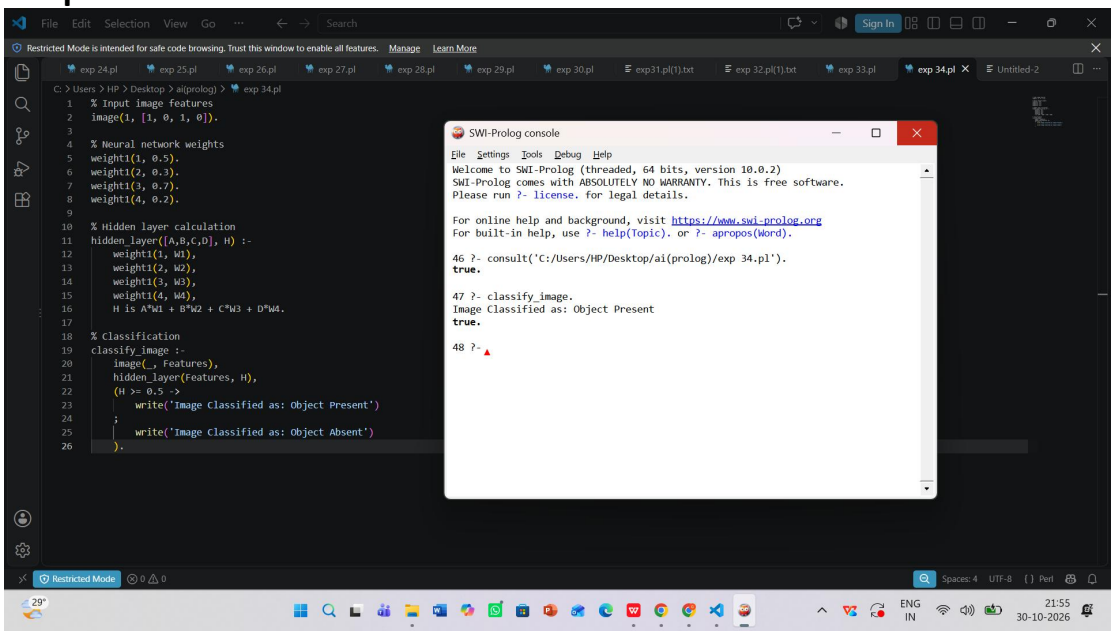
```
% Input image features
image(1, [1, 0, 1, 0]).

% Neural network weights
weight1(1, 0.5).
weight1(2, 0.3).
weight1(3, 0.7).
weight1(4, 0.2).

% Hidden layer calculation
hidden_layer([A,B,C,D], H) :-
   weight1(1, W1),
   weight1(2, W2),
   weight1(3, W3),
   weight1(4, W4),
   H is A*W1 + B*W2 + C*W3 + D*W4.

% Classification
classify_image :-
   image(_, Features),
   hidden_layer(Features, H),
   (H >= 0.5 ->
      write('Image Classified as: Object Present')
   ;
      write('Image Classified as: Object Absent')
   ).
```

Output:


File Edit Selection View Go
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exp 24.pl exp 25.pl exp 26.pl exp 27.pl exp 28.pl exp 29.pl exp 30.pl exp31.pl(1).txt exp 32.pl(1).txt exp 33.pl exp 34.pl Untitled-2
C: > Users > HP > Desktop > ai(prolog) > exp 34.pl
% Input image features
image(1, [1, 0, 1, 0]).
% Neural network weights
weight1(1, 0.5).
weight1(2, 0.3).
weight1(3, 0.7).
weight1(4, 0.2).
% Hidden layer calculation
hidden_layer([A,B,C,D], H) :-
weight1(1, W1),
weight1(2, W2),
weight1(3, W3),
weight1(4, W4),
H is A*W1 + B*W2 + C*W3 + D*W4.
% Classification
classify_image :-
image(_, Features),
hidden_layer(Features, H),
(H >= 0.5 ->
write('Image Classified as: Object Present')
;
write('Image Classified as: Object Absent')
).
SWI-Prolog console
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For built-in help, use ?- help(Topic). or ?- apropos(Word).
46 ?- consult('C:/Users/HP/Desktop/ai(prolog)/exp 34.pl').
true.
47 ?- classify_image.
Image Classified as: Object Present
true.
48 ?-
Restricted Mode
Spaces: 4 UTF-8 Perl
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